ST SPRINTS TALENT ANALYTICS

ADVANCED DATA ANALYSIS CAPSTONE

Analytical Report

Learner Performance, Data Quality, Feature Engineering, and Employer Skills Intelligence

Report Contents

Navigate through the analysis, findings, recommendations, and limitations.

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Executive Summary

A concise overview of the project and its analytical scope.

PROJECT OVERVIEW

This report summarizes learner performance, data quality, engineered features, and employer skill requirements. The analysis is descriptive and exploratory; results should be interpreted within the documented sample size, missing values, and skill-mapping limitations.

LEARNER RECORDS
24
Cleaned learner rows

TRAINING TRACKS
3
Distinct tracks

COHORTS
9
Distinct cohorts

EMPLOYER ROLES
50
Role records analyzed

DISTINCT SKILLS
60
Skills listed across roles

FEATURE COLUMNS
23
Engineered dataset columns

Dataset Overview

Structure and coverage of the available datasets.

Dataset Inventory

PROPERTY	VALUE
Clean learner dataset	sprints_cohort_clean.csv
Learner records	24
Learner columns	18
Feature dataset	sprints_cohort_features.csv
Feature records	24
Feature columns	23
Employer roles	50
Unique learner IDs	24

Learners by Training Track

TRAINING TRACK	LEARNERS
Advanced Data Analysis	13
AI & Machine Learning	6
Full Stack Web Development	5

Learners by Cohort

COHORT	LEARNERS
2023-Q1-AI	2
2023-Q1-DA	6
2023-Q1-FS	1
2023-Q2-AI	2
2023-Q2-DA	3
2023-Q2-FS	3
2023-Q3-AI	2
2023-Q3-DA	4
2023-Q3-FS	1

Learner records and employer role records are analyzed independently. The report does not assume that each learner corresponds to a particular employer role.

Data Quality & Cleaning

Cleaning decisions and remaining data-quality considerations.

Cleaning Summary

CLEANING REPORT FIELD	RECORDED VALUE
input_records	26
input_columns	18
structural_repairs	{'line_number': 26, 'learner_id': 'LRN-1023', 'original_fields': 19, 'final_fields': 18}
duplicate_rows_removed	2
invalid_engagement_values	{'learner_id': 'LRN-1002', 'original_value': 999.0, 'action': 'Converted to missing value'}, {'learner_id': 'LRN-1006', 'original_value': -20.0, 'action': 'Converted to missing value'}, {'learner_id': 'LRN-1012', 'original_value': 1500.0, 'action': 'Converted to missing value'}, {'learner_id': 'LRN-1020', 'original_value': -5.0, 'action': 'Converted to missing value'}
numeric_conversion_failures / engagement_score	0
numeric_conversion_failures / module_1_score	0
numeric_conversion_failures / module_2_score	0

CLEANING REPORT FIELD	RECORDED VALUE
numeric_conversion_failures / module_3_score	0
numeric_conversion_failures / module_4_score	0
numeric_conversion_failures / module_5_score	0
numeric_conversion_failures / module_6_score	0
numeric_conversion_failures / module_7_score	0
numeric_conversion_failures / module_8_score	0
numeric_conversion_failures / module_9_score	0
numeric_conversion_failures / module_10_score	0
numeric_conversion_failures / capstone_score	0
date_conversion_failures / enrollment_date	0
date_conversion_failures / graduation_date	0
missing_values_before_cleaning / learner_id	0
missing_values_before_cleaning / cohort_id	0
missing_values_before_cleaning / enrollment_date	0
missing_values_before_cleaning / graduation_date	2
missing_values_before_cleaning / demographic_group	0
missing_values_before_cleaning / track_name	0
missing_values_before_cleaning / engagement_score	1
missing_values_before_cleaning / module_1_score	0

CLEANING REPORT FIELD	RECORDED VALUE
missing_values_before_cleaning / module_2_score	1
missing_values_before_cleaning / module_3_score	1
missing_values_before_cleaning / module_4_score	0
missing_values_before_cleaning / module_5_score	2
missing_values_before_cleaning / module_6_score	3
missing_values_before_cleaning / module_7_score	2
missing_values_before_cleaning / module_8_score	3
missing_values_before_cleaning / module_9_score	3
missing_values_before_cleaning / module_10_score	3
missing_values_before_cleaning / capstone_score	3
missing_values_after_cleaning / learner_id	0
missing_values_after_cleaning / cohort_id	0
missing_values_after_cleaning / enrollment_date	0
missing_values_after_cleaning / graduation_date	2
missing_values_after_cleaning / demographic_group	0
missing_values_after_cleaning / track_name	0
missing_values_after_cleaning / engagement_score	5
missing_values_after_cleaning / module_1_score	0
missing_values_after_cleaning / module_2_score	1

CLEANING REPORT FIELD	RECORDED VALUE
missing_values_after_cleaning / module_3_score	1
missing_values_after_cleaning / module_4_score	0
missing_values_after_cleaning / module_5_score	2
missing_values_after_cleaning / module_6_score	3
missing_values_after_cleaning / module_7_score	2
missing_values_after_cleaning / module_8_score	3
missing_values_after_cleaning / module_9_score	3
missing_values_after_cleaning / module_10_score	3
missing_values_after_cleaning / capstone_score	3
output_records	24
output_columns	18

Score range assumption

The engagement score range of 0–100 is provisional and should be confirmed against the data dictionary or source documentation.

Missing Values by Column

COLUMN	MISSING VALUES	MISSING (%)
graduation_date	2	8.33
engagement_score	5	20.83
module_2_score	1	4.17
module_3_score	1	4.17
module_5_score	2	8.33
module_6_score	3	12.50
module_7_score	2	8.33
module_8_score	3	12.50

COLUMN	MISSING VALUES	MISSING (%)
module_9_score	3	12.50
module_10_score	3	12.50
capstone_score	3	12.50

Missing values are not zero

The cleaned dataset contains 28 missing cell values. Missing scores and dates are preserved where imputation has not been justified.

Cleaning Interpretation

Exact duplicate rows are removed while retaining one occurrence. Numeric conversion, engagement-range checks, date parsing, and structural CSV repairs are documented by the cleaning pipeline. The original source CSV is not overwritten.

Exploratory Data Analysis

Descriptive statistics and patterns in the learner data.

Assessment means and medians use available numeric values. The valid count is shown for each score so that differences in data coverage remain visible.

Assessment Score Summary

ASSESSMENT	MEAN	MEDIAN	AVAILABLE SCORES
Engagement Score	73.95	82	19
Module 1 Score	79.58	85	24
Module 2 Score	79.09	83	23
Module 3 Score	80.30	85	23
Module 4 Score	80.12	84.50	24
Module 5 Score	85.18	88.50	22
Module 6 Score	85.29	86	21
Module 7 Score	82.68	85.50	22
Module 8 Score	84.95	88	21
Module 9 Score	86.52	91	21
Module 10 Score	86.24	89	21
Capstone Score	86.05	88	21

Engagement and Capstone relationship

The Pearson correlation is 0.992, calculated from 16 complete records. Correlation does not establish causation, and the limited sample restricts generalization.

Interpretation

Differences between module averages are descriptive comparisons. Without information about assessment difficulty, grading criteria, learner progression, and comparable cohorts, they should not be treated as definitive evidence of improvement or decline.

Feature Engineering

Derived features, validation checks, and skill-mapping limitations.

Feature engineering derives additional variables from the cleaned learner dataset to support performance analysis and completeness checks.

Derived Features

FEATURE	DESCRIPTION	MINIMUM	MAXIMUM	MISSING
average_module_score	Mean of available module scores per learner.	41.25	97.10	0
module_score_count	Count of non-missing module scores per learner.	4	10	N/A
module_score_std	Sample standard deviation of available module scores per learner.	—	—	0
enrollment_year	Year extracted from enrollment_date.	—	—	0
days_to_graduation	Days between enrollment and graduation, when both dates are available and the duration is non-negative.	186	192	2

Feature Engineering Summary

VALIDATION CHECK	RESULT
Input records	24
Output records	24
Input columns	18
Output columns	23
Duplicate rows in output	0

Missing Values in Feature Dataset

FEATURE COLUMN	MISSING VALUES
graduation_date	2
engagement_score	5
module_2_score	1
module_3_score	1
module_5_score	2
module_6_score	3
module_7_score	2
module_8_score	3
module_9_score	3
module_10_score	3
capstone_score	3
days_to_graduation	2

Skill mapping limitation

Not performed. Module scores do not directly identify individual employer-required skills. Do not infer individual skill proficiency from general module scores without a validated assessment-to-skill mapping.

Feature Readiness Notes

POTENTIAL FEATURE	PURPOSE	VALIDATION REQUIRED
average_module_score	Mean of available module scores for each learner.	Confirm that module scores share the same scale.
module_score_count	Number of non-missing module scores for each learner.	Interpret missing scores as unavailable, not zero.
module_score_std	Standard deviation of available module scores for each learner.	Calculate only when enough module scores are present.
enrollment_year	Year extracted from enrollment_date.	Confirm date parsing before deriving the feature.
days_to_graduation	Difference between graduation_date and enrollment_date.	Only calculate when both dates are available.

Employer Requirements Analysis

Employer role distribution, thresholds, and requested skills.

Employer requirements are analyzed separately from learner assessment results. Skill frequency counts the number of role records listing each skill.

Company Tier Distribution

COMPANY TIER	NUMBER OF ROLES
Tier 1	20
Tier 2	19
Tier 3	11

Assessment Thresholds

THRESHOLD METRIC	VALUE
Valid threshold values	50
Minimum threshold	70
Median threshold	82
Mean threshold	83.06
Maximum threshold	97

Top 15 Most Requested Skills

RANK	SKILL	NUMBER OF ROLES
1	Python	50
2	Pandas	20
3	SQL	19
4	Data Visualization	9
5	Machine Learning	9
6	NumPy	8
7	Statistics	6
8	Matplotlib	4
9	Plotly	4
10	Tableau	4
11	A/B Testing	3
12	Deep Learning	3
13	Excel	3
14	Seaborn	3
15	Time Series Analysis	3

Skill frequency represents the number of role records listing a skill. It is not a direct measure of market-wide demand, hiring probability, salary impact, or learner proficiency.

▶ **View saved employer analysis details**

Key Findings

Evidence-based findings from the current analysis.

01 Learner distribution

The cleaned dataset contains 24 learner records across 3 training tracks and 9 cohorts. This describes the available dataset, not all learners.

02 Assessment performance

The average module score ranges from 41.25 to 97.10. These values describe this sample and are not universal performance benchmarks.

03 Most requested skill

Python appears in 50 employer role records out of 50. This reflects the provided requirements file.

04 Data completeness

Some assessment scores and graduation dates remain missing. Comparisons should account for valid observations for each metric.

05 Skill mapping

Learner module scores are not directly mapped to individual employer-required skills, so skill-level proficiency cannot be inferred.

Recommendations

Practical next steps to improve data quality and analytical value.

1 Prioritize data quality

Keep an audit trail for duplicate rows, invalid values, missing assessments, and structural CSV issues.

2 Validate score definitions

Confirm the engagement score range and assessment scoring rules with the dataset owner.

3 Investigate missing values

Ask why assessment scores and graduation dates are missing. Do not replace missing scores with zero without an approved rule.

4 Use employer requirements for curriculum review

Use frequently requested skills to guide curriculum discussions while considering role-specific needs and priority weights.

5 **Establish a validated skill mapping**

Document which assessments measure each skill and the evidence needed to assign proficiency.

6 **Expand the evidence base**

Repeat the analysis on larger cohorts and multiple time periods before making broad claims.

Limitations & Reproducibility

Interpretation boundaries and the artifacts used to build this report.

Limitations and Future Improvements

LIMITATION	INTERPRETATION
Sample size	The learner sample is limited; subgroup comparisons are exploratory.
Missing values	Incomplete assessment records can change the number of observations used in each statistic.
Score range assumption	The expected engagement score range should be confirmed with the data owner.
Skill mapping	Module scores are not a validated substitute for skill-level assessments.
Employer coverage	The employer file represents the roles included in this dataset, not necessarily the entire job market.
Further validation	Repeat the analysis with additional cohorts, documented scoring rules, and validated skill mappings.

Reproducibility: Input Artifacts

ARTIFACT	PROJECT PATH
Cleaned learner data	data/sprints_cohort_clean.csv
Engineered features	data/sprints_cohort_features.csv
Employer requirements	data/employer_partner_requirements.json
Cleaning report	reports/cleaning_report.json
EDA summary	reports/eda_summary.json
Employer analysis	reports/employer_requirements_summary.json
Feature readiness	reports/feature_engineering_readiness.json
Feature engineering report	reports/feature_engineering_report.json

Reproducibility note

The report is generated from saved JSON analysis reports and processed datasets. Rerun upstream analysis scripts before regenerating the HTML report.